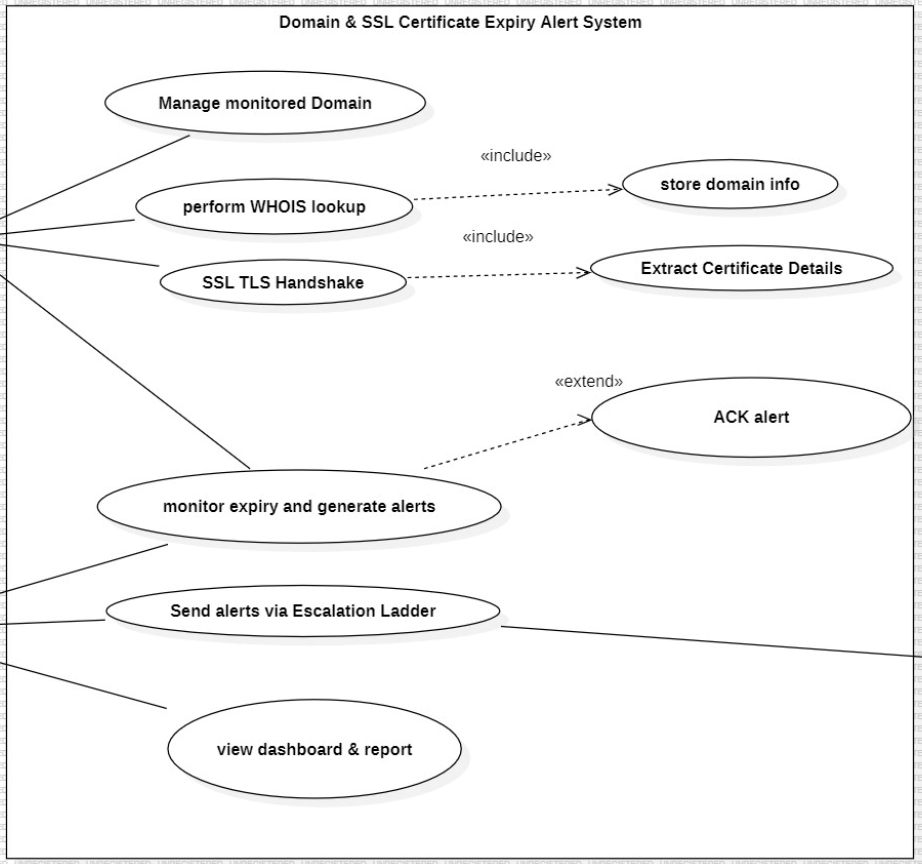
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Requirements Table:

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| **FR-001** | Functional | The system shall perform a daily WHOIS lookup for all monitored domains and store the retrieved domain registration details. | High | When the daily lookup runs, WHOIS data for **100% of the monitored domains** is retrieved and stored successfully. | Ensures up-to-date domain registration information is available for monitoring. |
| **FR-002** | Functional | The system shall perform a daily SSL/TLS handshake for all monitored domains and extract certificate details. | High | For every monitored domain, the system successfully retrieves the **certificate issuer, validity period, and expiry date**. | Required to monitor SSL/TLS certificate health and expiration. |
| **FR-003** | Functional | The system shall trigger alerts at **30, 15, and 3 days** before domain or certificate expiry. | High | An alert is generated when a monitored domain or certificate reaches **30, 15, or 3 days** before expiry. | Provides timely warning to take preventive action. |
| **FR-004** | Functional | The system shall send alerts to configured stakeholders through an escalation ladder until the alert is acknowledged. | High | Notifications are sent to configured stakeholders in the defined escalation order until acknowledgement is received. | Ensures critical expiry alerts reach the appropriate personnel on time. |
| **FR-005** | Functional | The system shall provide a dashboard displaying the current status of monitored domains and SSL certificates. | Medium | The dashboard displays the status of monitored domains, upcoming expirations, and the latest scan time. | Provides quick visibility into the overall monitoring status. |
| **NFR-001** | Non-Functional (Performance) | The monitoring engine shall scan a list of **1,000 domains and SSL endpoints in under 3 minutes**. | High | Benchmark testing confirms that 1,000 domains and SSL endpoints are scanned within **3 minutes**. | Ensures efficient monitoring at scale. |
| **NFR-002** | Non-Functional (Security) | The system shall protect monitored domain and certificate information during transmission and storage using appropriate encryption. | High | Data transmission uses **TLS 1.2 or higher**, and sensitive stored information is encrypted at rest. | Protects monitoring data from unauthorized access and interception. |

| Field | Content |
| --- | --- |
| **Goal** | Monitor Domain & SSL Certificate Expiry and Generate Alerts |
| **Actor** | SysAdmin with help from Security Officer and Notification Service |
| **Preconditions** | 1. Monitored domains are configured in the system.2. WHOIS and SSL/TLS monitoring services are available.3. Alert recipients and escalation levels are configured. |
| **Postconditions** | 1. Domain and certificate information is updated.2. Expiring domains/certificates are identified.3. Required alerts are generated and sent.4. Monitoring status is updated on the dashboard. |
| **Main Flow** | **1.** SysAdmin configures the domains to be monitored.**2.** The system initiates the daily WHOIS lookup for all monitored domains.**3.** The system retrieves and stores domain registration information.**4.** The system performs an SSL/TLS handshake for each monitored domain.**5.** The system extracts certificate details including issuer, validity period, and expiry date.**6.** The system checks whether any domain or certificate is expiring in 30, 15, or 3 days.**7.** The system generates alerts for the identified expiring items.**8.** The system sends alerts to stakeholders through the configured escalation ladder.**9.** The stakeholder acknowledges the alert, if required.**10.** The system records the acknowledgement status.**11.** The system updates the dashboard with the latest monitoring status.**12.** Monitoring process is completed. |
| **Alternate Flow** | **Domain Unreachable / SSL Handshake FailureAt Step 4**, if the system cannot connect to a monitored domain or the SSL/TLS handshake fails:**4a.** The system records the connection or handshake failure.**4b.** The system marks the domain as **Unreachable / Scan Failed**.**4c.** The system generates an alert for the failed monitoring attempt.**4d.** The system continues monitoring the remaining domains. |

Use-Case Flow Specification:


Domain & SSL Certificate Expiry Alert System
Manage monitored Domain
perform WHOIS lookup
«include»
store domain info
SSL TLS Handshake
«include»
Extract Certificate Details
Admin
«extend»
ACK alert
monitor expiry and generate alerts
Send alerts via Escalation Ladder
Security officer
Notification service
view dashboard & report